

§13 Vector Functions

§13.1 Vector Functions

Homework : 3,14,19-24,35

Functions :

- (i) $f : R \rightarrow R$
- (ii) $f : R \rightarrow R^m$
- (iii) $f : R^n \rightarrow R$
- (iv) $f : R^n \rightarrow R^m$

(i)型的函數之微分和積分是我們之前微積分課程學的東西。(ii)和(iii)型函數的微分和積分是這學期剩下的時間要學習的。至於(iv)型函數的微分在高微的課程會學到。

(1) Vector Functions or Vector-valued Functions

(ii)型函數即稱 Vector Functions.

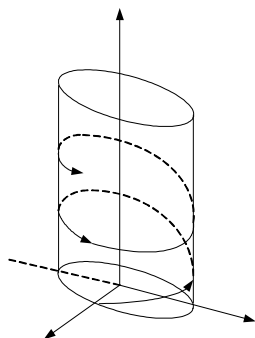
如 $r(t) = \langle f(t), g(t), h(t) \rangle = f(t)i + g(t)j + h(t)k$, 其中 f, g, h 是實函數.

(2) Space curve : If f, g, h are continuous real-valued functions, then the orbit of

$r(t) = \langle f(t), g(t), h(t) \rangle$ is called a space curve.

Example 1 : $r(t) = \langle 1+t, 2+5t, -1+6t \rangle$ 此函數圖形為直線.

Example 2 : $r(t) = \langle \cos t, \sin t, t \rangle$



$$\begin{aligned}x^2 + y^2 &= 1 \\ z &= t\end{aligned}$$

此函數圖形稱helix(螺旋).

Example 3 : Find a vector function that represents the curve of intersection of the

two surfaces : the cone $z = \sqrt{x^2 + y^2}$ and the plane $z = 1 + y$.

Solution :

$$\begin{aligned}(1+y)^2 &= x^2 + y^2 \\ \Rightarrow x^2 &= 1 + 2y \\ r(t) &= \left(t, \frac{t^2 - 1}{2}, \frac{t^2 + 1}{2} \right)\end{aligned}$$

Example 4 : 19-24 of P.855.

Match the parametric equations with the graphs (labeled I-VI).

Give reasons for your choices.

19. $x = \cos 4t$, $y = t$, $z = \sin 4t$

沿著 y 軸而上的 helix. $\Rightarrow$ VI

20. $x = t$, $y = t^2$, $z = e^{-t}$

y 、 z 皆為正， x 愈大， z 愈小. $\Rightarrow$ II

21. $x = t$, $y = \frac{1}{1+t^2}$, $z = t^2$

y 、 z 皆為正， x 愈大， z 愈大. $\Rightarrow$ IV

22. $z = e^{-t} \cos 10t$, $y = e^{-t} \sin 10t$, $x = e^{-t}$

$x^2 + y^2 = z^2$ (cone, 軸為 z -axis), 且 z 皆為正. $\Rightarrow$ I

23. $x = \cos t$, $y = \sin t$, $z = \sin 5t$

$x^2 + y^2 = 1$ x, y, z 皆為週期. $\Rightarrow$ V

24. $x = \cos t$, $y = \sin t$, $z = \ln t$

$x^2 + y^2 = 1$ 且 $z \rightarrow \infty$ as $t \rightarrow 0^+$. $\Rightarrow$ III